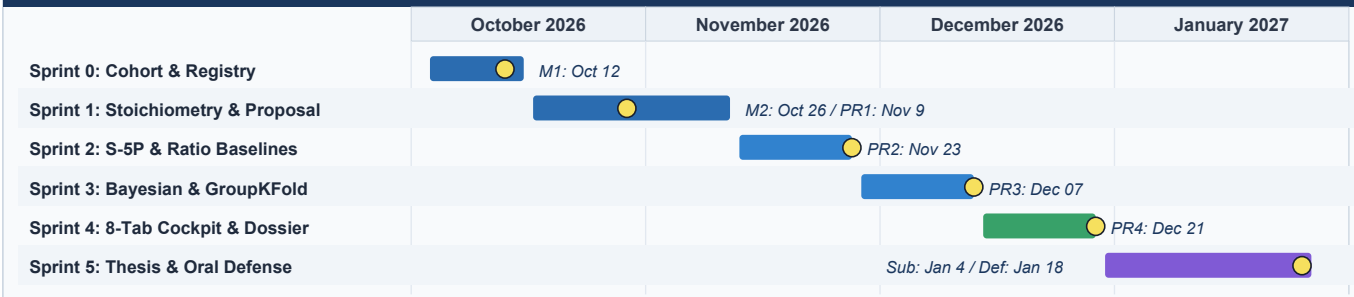
VeriCBAM: Capstone Project Work Plan, Risk Matrix & Execution Roadmap

A Data-Driven Decision-Support Tool for Screening Carbon Border (CBAM) Emissions Reports

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VeriCBAM Execution Gantt Chart & Milestone Timeline (Winter Semester 2026/2027)



1. Project Execution Schedule & Academic Milestones (Winter 2026/2027)

Phase / Sprint	Timeline	Key Activities & Tasks	Primary Deliverable / Milestone
Milestone 1 & Sprint 0	Oct 03 – Oct 14, 2026	Define project scope; collect and clean historical emissions data for 30 European cement and steel plants (177 annual records) and plant capacity data.	M1: Topic Approval (Oct 12, 2026) & Cleaned Facility Dataset
Milestone 2 & Sprint 1	Oct 15 – Nov 09, 2026	Finalize 1-page proposal; implement Check 1 (Physical Minimums) and Check 2 (Production Technology) for cement and steel plants.	M2: Proposal Approval (Oct 26, 2026) & Progress Report 1 (Nov 09)
Sprint 2	Nov 10 – Nov 23, 2026	Implement Check 3 (Historical Trends), Check 4 (NO_x vs. CO_2 Pollutant Ratios), and Check 5 (Copernicus Sentinel-5P Satellite NO_2 extraction).	M4: Progress Report 2 (Nov 23, 2026) & 5-Check Data Pipeline
Sprint 3	Nov 24 – Dec 07, 2026	Build the Bayesian Risk Fusion model and 5-fold Cross-Validated Logistic Regression model; run evaluation across the 325-case benchmark.	M5: Progress Report 3 (Dec 07, 2026) & Model Comparison Results
Sprint 4	Dec 08 – Dec 21, 2026	Build the interactive Streamlit Web Dashboard, visual map inspector, 1-click Audit Dossier generator, and automated unit tests (`pytest`).	M6: Progress Report 4 (Dec 21, 2026) & Live Web Dashboard
Sprint 5 & Defense	Dec 22 – Jan 18, 2027	Write and proofread the final Master Thesis manuscript, verify code reproducibility, and prepare the final defense presentation.	M7: Final Submission (Jan 04, 2027) & M8: Defense (Jan 18, 2027)

2. Project Risk Management Matrix

Identified Risk	Severity	Likelihood	Impact on Project	Mitigation Strategy
Cloud Cover Blocking Satellite Data	High	High	Missing satellite observations over northern Europe during winter.	Weight satellite evidence by cloud-free visibility; if data is too cloudy, automatically reduce its weight or output "Insufficient Evidence".
Shared Factory Energy / Off-Site Gases	High	Medium	Some steel plants send furnace gas to nearby power plants, making their own stack emissions look artificially low.	Flag factories with off-site gas transfers in the plant registry so they are not falsely accused by the physical minimum check.
Missing Data in Public Registries	Medium	Medium	A few factories have missing reporting years in public databases.	Do not invent or guess missing years; leave them blank (`NaN`) and rely on the remaining available checks.
Data Leakage in Machine Learning	Critical	Low	Testing a model on the same factory it was trained on would give unrealistic accuracy.	Use Grouped 5-Fold Cross-Validation (`GroupKFold` by facility), ensuring all years of a factory are kept strictly in either training or testing.
Lack of Public Fraud Examples	High	Medium	Real fraudulent CBAM reports are legally confidential and not publicly available.	Create a transparent 325-case test benchmark by combining 177 real compliant reports with controlled under-reporting scenarios (5% to 50%).

3. Expected Deliverables & Resources Required

• Core Deliverables:

1. Python Data & Modeling Package (`src/`): Clean, documented code implementing the five evidence checks, Bayesian risk fusion, and cross-validated logistic regression.
2. Interactive Streamlit Web Application (`src/ui/app.py`): A visual dashboard allowing users to explore the 30 facilities on a map, test "what-if" reporting scenarios, view the risk breakdown, and export a 1-page Audit Dossier.
3. Automated Test & Reproduction Suite (`tests/`, `scripts/`): Unit tests verifying code reliability and a single script to reproduce all thesis tables and charts.

• Resources & Supervision: Runs on a standard laptop using open-source Python libraries (`pandas`, `scikit-learn`, `streamlit`, `plotly`, `pytest`) and free public datasets (EEA E-PRTR, Global Energy Monitor, Copernicus Sentinel-5P), reviewed bi-weekly with Dr. Humera Noor.

4. Future Extension Roadmap (Stage 2 Horizon)

Once the Stage 1 system is complete, the framework can be extended with higher-resolution satellite data to estimate factory activity directly from space:

1. Thermal Infrared Satellite Data (Landsat / NASA VIIRS): Detecting heat directly from cement kilns and steel blast furnaces to verify how many months per year a plant was actively running.
2. High-Resolution Optical Imagery (PlanetScope): Observing factory rail yards and raw material stockpiles to confirm physical production activity independently of self-reported numbers.